6 Fig. 6.1 shows the current-voltage (*I*-*V*) characteristics of a resistor and a diode.

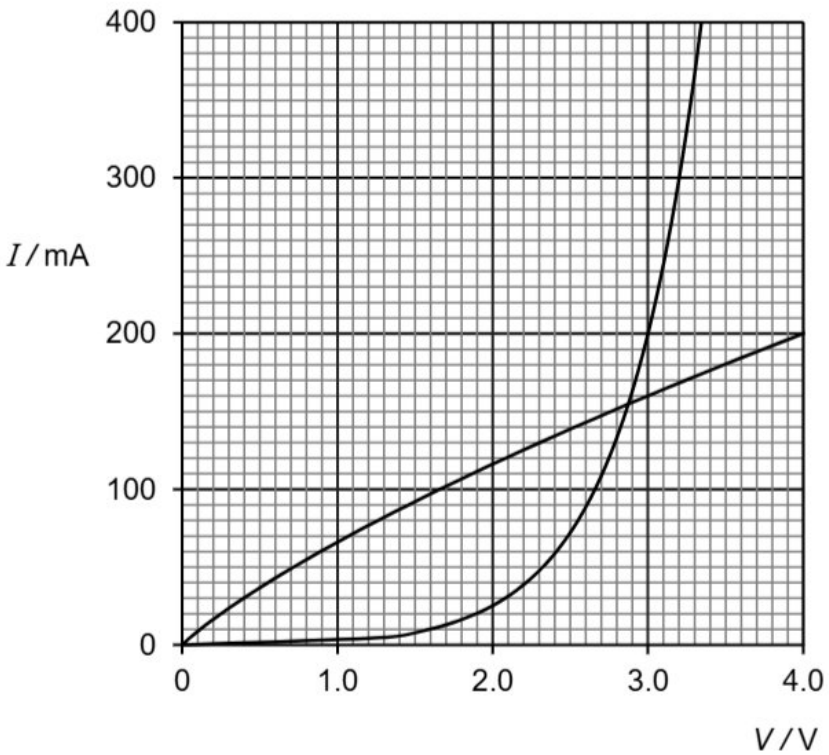


Fig. 6.1

When the resistor and diode are connected in series with a battery of negligible internal resistance, the current in the circuit is 200 mA.

(a) Determine

(i) the e.m.f. of the battery,

e.m.f. = ... V [2]

(ii) the resistance of the diode.

resistance = ... Ω [1]

(b) (i) On Fig. 6.1, sketch the I-V characteristic of a negative temperature coefficient thermistor. Exact values are not required. [1]

(ii) Explain, in microscopic terms, the difference in the I-V characteristics between the thermistor and the resistor.

.....

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..... [1]

[Total: 5]